

Growing-Depth Lower-Sideband Phase Incompatibility

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Abstract

We extend the two-coordinate phase obstruction of RH-349 to a triangular lower-even ladder. Let $J_k \rightarrow \infty$, $J_k = o(k)$, and $m_{k,j} = k - j$ for $2 \leq j \leq J_k$. The exact direct coefficient is

$$p_{k,j} = Y_{k,j} + \mathcal{P}_{k,j} - S_{k,j}.$$

Writing $H_m = mR^{-2m}$, $x = (\beta R)^2 > 1$, and $a_k = C_* C_M \lambda^{\eta_k - 2}$, we prove the uniform deterministic laws

$$\sup_{2 \leq j \leq J_k} \left| \frac{C_M S_{k,j}}{2H_{m_{k,j}} x^{m_{k,j}}} - 1 \right| \rightarrow 0,$$

$$\sup_{2 \leq j \leq J_k} \left| \frac{C_M \mathcal{P}_{k,j}}{2H_{m_{k,j}} x^{m_{k,j}}} - a_k \lambda^{2-j} \right| \rightarrow 0.$$

For a fixed depth $J \geq 3$, the relative phase mismatch has exact minimax value

$$\inf_{a > 0} \max_{2 \leq j \leq J} |a \lambda^{2-j} - 1| = \frac{\lambda^{J-2} - 1}{\lambda^{J-2} + 1}.$$

For the physical prefix weighting, put $F_N(a) = \sum_{r=0}^N x^{-r} |a \lambda^{-r} - 1|$. Since $x \lambda = (28/17)^2 > 2$, its unique minimizer is $a = 1$ and

$$\inf_{a > 0} F_N(a) = \frac{1 - x^{-N}}{x - 1} - \frac{1 - (x \lambda)^{-N}}{x \lambda - 1}.$$

Under the explicit, presently unproved actual aggregate hypothesis

$$x^{-(k-2)} \sum_{j=2}^{J_k} \frac{|Y_{k,j}|}{2H_{m_{k,j}}} \rightarrow 0,$$

the selected direct subprefix therefore has a strictly positive normalized liminf and diverges exponentially. This is a conditional growing-depth theorem, not unconditional physical nonclosure. Odd orders, upper-alias orders, the full off-alias aggregate, determinant gluing, and Gates A–E remain open. No Riemann-hypothesis conclusion is made.

1 The triangular lower-even ledger

Use the physical clock and bounded phase

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \eta_k = k - \frac{\log(1/\sigma)}{2 \log \lambda}. \quad (1)$$

Let J_k be integer-valued with

$$J_k \longrightarrow \infty, \quad J_k = o(k), \quad J_k \geq 3, \quad (2)$$

and set, for $2 \leq j \leq J_k$,

$$m_{k,j} = k - j, \quad n_{k,j} = 2m_{k,j}, \quad H_m = mR^{-2m}, \quad x = (\beta R)^2. \quad (3)$$

Because $k - J_k \rightarrow \infty$, every selected $m_{k,j}$ eventually belongs to the RH-348 ladder I_k [3].

Abbreviate

$$p_{k,j} = p_{\sigma,k,2m_{k,j}}, \quad \mathcal{P}_{k,j} = \mathcal{P}_{\sigma,2m_{k,j}}.$$

The simultaneous complete-orbit extraction gives, exactly,

$$\boxed{p_{k,j} = Y_{k,j} + \mathcal{P}_{k,j} - S_{k,j}}, \quad (4)$$

where

$$Y_{k,j} = \mathcal{T}_{k,m_{k,j}}^{\text{rest}} - d_{\sigma,k,2m_{k,j}}, \quad S_{k,j} = F_{m_{k,j}}^{\text{orb}} + \mathcal{A}_{k,2m_{k,j}}. \quad (5)$$

Here

$$F_m^{\text{orb}} = 2mG_m, \quad G_m = \frac{r_H^{-2m}}{1 + |M_m|}, \quad (6)$$

$$\mathcal{A}_{k,2m} = 2(\beta^{2m} - \beta_k^{2m}), \quad \beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}. \quad (7)$$

The coefficient p is direct. The actual signed term $Y_{k,j}$ retains both the orbit-free trace remainder and the noisy-head/counterloop defect; no full-trace/direct substitution is made [4, 7].

2 Uniform deterministic and parity laws

Theorem 2.1 (Growing-window scalar uniformity). *Under (1)–(2), put*

$$a_k = C_* C_M \lambda^{\eta_k - 2}. \quad (8)$$

Then

$$\boxed{\sup_{2 \leq j \leq J_k} \left| \frac{C_M S_{k,j}}{2H_{m_{k,j}} x^{m_{k,j}}} - 1 \right| \rightarrow 0}, \quad (9)$$

and

$$\boxed{\sup_{2 \leq j \leq J_k} \left| \frac{C_M \mathcal{P}_{k,j}}{2H_{m_{k,j}} x^{m_{k,j}}} - a_k \lambda^{2-j} \right| \rightarrow 0}. \quad (10)$$

These are uniform deterministic/scalar packet laws. They contain no estimate for the actual signed $Y_{k,j}$.

Proof. Write

$$|M_m| = C_M \lambda^m \{1 + \varepsilon_m\}, \quad \varepsilon_m \rightarrow 0.$$

Since $m_{k,j} \geq k - J_k \rightarrow \infty$, $\sup_{2 \leq j \leq J_k} |\varepsilon_{m_{k,j}}| \rightarrow 0$. Thus, uniformly on the selected window,

$$F_{m_{k,j}}^{\text{orb}} = \frac{2m_{k,j}}{C_M} \beta^{2m_{k,j}} \{1 + o(1)\}. \quad (11)$$

The finite-radius law

$$\beta_k = \beta \exp \left[-\frac{\log C_M}{2k} + o(k^{-1}) \right]$$

and the RH-348 exponential bound give, uniformly for $m \leq k$,

$$|\mathcal{A}_{k,2m}| \leq C \frac{m}{k} \beta^{2m}.$$

After division by (11), the radial term is $O(k^{-1})$ uniformly. Hence $S_{k,j}$ has the same uniform leading law as the orbit term. Since $2H_m x^m = 2m\beta^{2m}$, this proves (9) [1, 3].

For every even order, the exact parity formula is

$$\mathcal{P}_{\sigma,2m} = r_H^{-2m} \{1 - (1 - \delta_\sigma)^{2m}\}, \quad \delta_\sigma = C_* \sqrt{\sigma} \{1 + o(1)\}.$$

The all-order Taylor remainder in RH-326 yields

$$\mathcal{P}_{\sigma,2m} = 2m C_* \sqrt{\sigma} r_H^{-2m} \{1 + o(1)\}$$

uniformly for $m \leq k$, because $k\delta_\sigma \rightarrow 0$. Therefore

$$\frac{C_M \mathcal{P}_{k,j}}{2H_{m_{k,j}} x^{m_{k,j}}} = C_* C_M \sqrt{\sigma} \lambda^{m_{k,j}} + o(1) = a_k \lambda^{2-j} + o(1)$$

uniformly in j . Boundedness of (1) makes a_k bounded and justifies the final uniform absolute error [2, 5]. $\square$

The growing-depth conclusion is not inferred from finitely many fixed- j limits. It follows from a tail-uniform multiplier asymptotic, an all-order parity remainder, and the explicit radial bound on one common window.

3 Finite-depth relative minimax

Proposition 3.1 (Exact J -coordinate relative minimax). *For every integer $J \geq 3$ and $\lambda > 1$,*

$$\boxed{\inf_{a>0} \max_{2 \leq j \leq J} |a\lambda^{2-j} - 1| = \frac{\lambda^{J-2} - 1}{\lambda^{J-2} + 1}.} \quad (12)$$

The unique minimizer is

$$a_J^{\text{rel}} = \frac{2\lambda^{J-2}}{\lambda^{J-2} + 1}. \quad (13)$$

In particular, the optimal largest relative mismatch tends to one as $J \rightarrow \infty$.

Proof. The values $a\lambda^{2-j}$ decrease with j . Convexity of $t \mapsto |t - 1|$ implies that the largest mismatch occurs at one of the two endpoints $j = 2$ and $j = J$. Hence the problem is

$$\inf_{a>0} \max\{|a - 1|, |a\lambda^{2-J} - 1|\}.$$

At the minimizer the two endpoint errors are equal with opposite signs:

$$a - 1 = 1 - a\lambda^{2-J}.$$

Solving gives (13) and substitution gives (12). The endpoint regimes are monotone, so the minimizer is unique. $\square$

4 Exact physical weighted minimax

For $N \geq 1$, define

$$F_N(a) = \sum_{r=0}^N x^{-r} |a\lambda^{-r} - 1|, \quad a > 0. \quad (14)$$

The physical constants satisfy the exact identity

$$x\lambda = \left(\frac{R}{r_H}\right)^2 = \left(\frac{28}{17}\right)^2 > 2. \quad (15)$$

This rational dominance is the exact form of the superunit certificate used in the physical Hardy normalization [6].

Theorem 4.1 (Weighted-median phase law). *For every $N \geq 1$, the unique minimizer of F_N is $a = 1$, and*

$$\inf_{a>0} F_N(a) = A_N := \frac{1 - x^{-N}}{x - 1} - \frac{1 - (x\lambda)^{-N}}{x\lambda - 1}. \quad (16)$$

Moreover,

$$A_N \nearrow A_\infty := \frac{1}{x - 1} - \frac{1}{x\lambda - 1} > 0. \quad (17)$$

Proof. Rewrite each summand as

$$x^{-r} |a\lambda^{-r} - 1| = (x\lambda)^{-r} |a - \lambda^r|.$$

Thus F_N is a weighted absolute-deviation functional with knots $1, \lambda, \dots, \lambda^N$ and weights $1, (x\lambda)^{-1}, \dots, (x\lambda)^{-N}$. By (15),

$$\sum_{r=1}^N (x\lambda)^{-r} < \sum_{r=1}^{\infty} (x\lambda)^{-r} < 1.$$

The first knot therefore carries more weight than all later knots combined, so the unique weighted median is $a = 1$. Evaluation gives

$$F_N(1) = \sum_{r=1}^N x^{-r} (1 - \lambda^{-r}),$$

and the two geometric sums yield (16). Every added summand is positive, and passage to the two infinite geometric series proves (17). $\square$

With the alternative bottom-sideband normalization, theorem 4.1 is equivalently

$$\inf_{a>0} \sum_{j=2}^J x^{J-j} |a\lambda^{2-j} - 1| = \frac{x^{J-2} - 1}{x - 1} - \frac{x^{J-2} - \lambda^{2-J}}{x\lambda - 1}.$$

At $J = 3$ this reduces to the RH-349 constant $1 - \lambda^{-1}$ [7].

5 Conditional growing-depth direct-prefix obstruction

Put

$$W_{k,j} = \frac{|p_{k,j}|}{2H_{m_{k,j}}} = \frac{|p_{k,j}| R^{2m_{k,j}}}{2m_{k,j}}. \quad (18)$$

The actual aggregate remainder scale on the selected triangular window is

$$\mathcal{Y}_k := \frac{1}{x^{k-2}} \sum_{j=2}^{J_k} \frac{|Y_{k,j}|}{2H_{m_{k,j}}}. \quad (19)$$

Theorem 5.1 (Conditional growing-depth divergence). *Suppose the actual physical coefficients satisfy the explicit hypothesis*

$$\boxed{\mathcal{Y}_k \longrightarrow 0.} \quad (20)$$

Then, with $N_k = J_k - 2$,

$$\boxed{\frac{1}{x^{k-2}} \sum_{j=2}^{J_k} W_{k,j} = \frac{F_{N_k}(a_k)}{C_M} + o(1).} \quad (21)$$

Consequently,

$$\liminf_{k \rightarrow \infty} \frac{1}{x^{k-2}} \sum_{j=2}^{J_k} W_{k,j} \geq \frac{1}{C_M} \left(\frac{1}{x-1} - \frac{1}{x\lambda-1} \right) > 0. \quad (22)$$

The selected lower-even direct subprefix therefore diverges exponentially.

Proof. For each j , the elementary inequality

$$||\mathcal{P}_{k,j} - S_{k,j} + Y_{k,j}| - |\mathcal{P}_{k,j} - S_{k,j}|| \leq |Y_{k,j}|$$

shows that hypothesis (20) permits deletion of the actual Y aggregate at the normalization in (21). By theorem 2.1, uniformly in j ,

$$\frac{|\mathcal{P}_{k,j} - S_{k,j}|}{2H_{m_{k,j}}x^{m_{k,j}}} = \frac{|a_k\lambda^{2-j} - 1|}{C_M} + o(1).$$

Since $x^{m_{k,j}}/x^{k-2} = x^{2-j}$ and $\sum_{j=2}^{\infty} x^{2-j} < \infty$, summing the uniform error gives $o(1)$. Reindexing by $r = j - 2$ proves (21). Apply theorem 4.1 with $N = N_k \rightarrow \infty$ to obtain (22). $\square$

Corollary 5.2 (A stronger coefficientwise sufficient hypothesis). *If*

$$\max_{2 \leq j \leq J_k} \frac{|Y_{k,j}|}{H_{m_{k,j}}} \longrightarrow 0, \quad (23)$$

then (20) holds and hence so do (21)–(22).

Proof. The left side of (19) is at most

$$\frac{J_k}{2x^{k-2}} \max_{2 \leq j \leq J_k} \frac{|Y_{k,j}|}{H_{m_{k,j}}},$$

which tends to zero because $J_k = o(k)$ while $x > 1$. $\square$

Neither (20) nor (23) is proved in the repository. They concern the actual signed orbit-free trace remainder and noisy-head/counterloop defect. The theorem is therefore conditional and does not establish unconditional full-prefix or E_{off} nonclosure.

6 Executable protocol and route boundary

The artifact evaluates the exact finite multiplier, orbit, radial, and leading-law parity formulas for several pairs (k, J) with $J = \lfloor \sqrt{k} \rfloor$. It uses the scalar fixture $a_k = 1$ and $Y_{k,j} = 0$ solely to reproduce the deterministic formulas. It also checks the exact relative minimax, the weighted-median identity, monotone convergence of A_N , and the identity $x\lambda = (28/17)^2$. The rows are not interval certificates, asymptotic evidence, or observations of the actual Y aggregate.

RH-350 proves a genuinely growing-depth deterministic uniformity theorem, an exact family of finite-depth minimax laws, and a conditional selected direct-subprefix obstruction. It does not estimate the actual signed remainder, control odd or upper-alias orders, decide the full E_{off} aggregate, or activate RH-288. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

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